Polyphony J. S. Bruna

PhD Candidate · Cognitive and Information Sciences

University of California, Merced

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Education

Research Interests: Neural Networks, Complex & Dynamical Systems, Collective Intelligence, Joint Action, Alignment

University of California, Merced — *Merced, CA*

PhD Cognitive and Information Sciences — *Expected 2027*

- Committee: Christopher T. Kello (chair), Michael J. Spivey, and Jeff Yoshimi
- Master's thesis (*en route*): "Synergetic Mechanisms Underlying the Emergence of Alignment in Human Dyadic Interaction"

University of California, Los Angeles — *Los Angeles, CA*

Graduate Fellow, Institute for Pure and Applied Mathematics — *Fall 2024*

- Program: Mathematics of Intelligences

Vassar College — *Poughkeepsie, NY*

BA Cognitive Science and Philosophy (Minor: Japanese) — *2018 - 2022*

- Overall GPA: 3.99/4.00
- Honors: *Phi Beta Kappa, Sigma Xi, Psi Chi, cum laude generali et cum laude in materia subiecta*
- Thesis: "Leveraging Surprise in a Recurrent Neural Network to (de)Construct Morphological Complexity in Japanese" (Advisors: Jan Andrews and Josh de Leeuw)

Professional Experience

Memberships

2022-cont. **Phi Beta Kappa**, Academic Honor Society
2022-cont. **Sigma Xi**, The Scientific Research Honor Society
2021-cont. **Psi Chi**, International Honor Society in Psychology

Positions

2024, 2026 **Humane Studies Fellow**, Institute for Humane Studies, George Mason University
2022-2023 **Adolph Sutro Fellow**, Vassar College
2021-2022 **Intern**, Cognitive Science, Vassar College
2021 **Senior Researcher**, Undergraduate Research Summer Institute, Vassar College
2019, 2020 **Research Fellow**, Undergraduate Research Summer Institute, Vassar College
2018-2021 **Research Assistant**, Cognitive Science, Vassar College

Publications

** Denotes co-principal authorship*

In Preparation

Bruna, P., Dale, R., Spivey, M., & Kello, C. (n.d.). Collective advantage and efficiency of group interaction in coupled echo state networks [under review at *PNAS*]. https://arxiv.org/abs/2610.17506

Bruna, P. & Gyllingberg, L. (2025). *Cognition without neurons: modeling anticipation in a basal reservoir computer*. https://arxiv.org/abs/2505.02114

Published

Kello, C., **Bruna, P.**, & Thao, K. (2026). Contextual assembly of lexical functions in large language models. *Behavior Research Methods 58*(19). https://doi.org/10.3758/s13428-025-02898-7

Bruna, P. and Kello, C. (2025). Least effort and alignment in task-oriented communication. *Cognitive Science, 49*, Article e70062. https://doi.org/10.1111/cogs.70062

de Leeuw, J. R., Andrews, J., Barney, L., Bigler, M., **Bruna, P.**, Chen, Y., Cherry, R., Dowie, D. R., Forbes, E., Haffey, B., Hu, X., Jaklitsch, M., Leopold, N., Lewis, C., MacDonald, D., McShaffrey, C., Nakayama, K., Olstad, W., Peng, R., … Zhang, L. (2021). Words may jump-start meaning more than vision: a non-replication of early ERP effects in Boutonnet and Lupyan (2015). *Collabra: Psychology, 7*(1). https://doi.org/10.1525/collabra.29763

Proceedings

Bruna, P., Dale, R., Spivey, M., & Kello, C. (2026). Collective decision-making in coupled echo state networks. *Proceedings of the Annual Meeting of the Cognitive Science Society, 48.* https://escholarship.org/uc/item/4p27s372

Nosrati, S., **Bruna, P.**, Spivey, M., & Marghetis, T. (2026). Spatial navigation at the extremes: The collective accomplishment of high-speed, high-risk navigation in Rally Racing. *Proceedings of the Annual Meeting of the Cognitive Science Society, 48.* https://escholarship.org/uc/item/0kr3d6pw

Kello, C. & **Bruna, P.** (2024). Emergent mental lexicon functions in ChatGPT. *Proceedings of the Annual Meeting of the Cognitive Science Society, 46.* https://escholarship.org/uc/item/5m9098b5

*Rane, S., ***Bruna, P.**, Sucholutsky, I., Kello, C., & Griffiths, T. (2023). Concept alignment. *1st NeurIPS Workshop on AI meets Moral Philosophy and Moral Psychology (MP2).* https://openreview.net/pdf?id=D3GCdfJL4J

Miscellaneous

Yoshimi, J., Beckmann, P., Udell, D., **Bruna, P.**, & Meyer, T. (2025). Transformer architectures and LLMs. In Yoshimi, J., Udell, D., Tosi, Z., Hotton, S., Beckmann, P., Cain, E., Gordon, C., Noelle, D. C., **Bruna, P.**, & Meyer, T. (Eds.), *Neural Networks in Cognitive Science*. https://downloads.jeffyoshimi.net/NeuralNetworksCogsci.pdf

Abbas, M., Alharthi, D., Balbas, J., Beckelhymer, D., Beylkin, G., **Bruna, P.**, Cartmill, E., Chormai, P., Corea, F., Dai, X., Das, P., Dipoppa, M., Eberle, O., El Hady, A., Evans, J., Feghhi, E., Flack, J., Flores, C., Foster, J., … Zhu, Y. (2025). White Paper: "Mathematics of Intelligences." *Institute for Pure and Applied Mathematics*. https://www.ipam.ucla.edu/reports

Bruna, P. (2019). A review of Bryan W. Van Norden, Taking Back Philosophy. *Vassar College Journal of Philosophy, 6.* https://www.vassar.edu/sites/default/files/2021-08/philosophy-journal-spring-2019.pdf

Presentations

Invited Talks

Fall 2026. *Emergent Performance and Collective Efficiency in Distributed, Coupled Echo State Networks*. Centre for Cognition, Donders Institute for Brain, Cognition, and Behaviour. Nijmegen, Netherlands.

Summer 2026. *Collective Decision-Making in Coupled Echo State Networks*. La Selva Conference on Language and Emergence. Santa Cruz, CA.

Spring 2026. *Collective Decision-Making in Coupled Echo State Networks*. Modeling the Social Mind(s): Theory-Based Computational Models (Symposium), The Society for Personality and Social Psychology Annual Convention. Chicago, IL.

Spring 2026. *Collective Decision-Making in Coupled Echo State Networks*. Stanford University, CA.

Fall 2025. *Collective Decision-Making in Coupled Echo State Networks*. December Graduate Conference, Institute for Humane Studies, George Mason University. Fairfax, VA.

Fall 2024. *Least Effort and Alignment in Task-Oriented Communication*. Vassar College. Poughkeepsie, NY.

Contributed Talks

Bruna, P. 2026. *Collective Decision-Making in Coupled Echo State Networks*. 48th Annual Conference of the Cognitive Science Society, Rio de Janeiro, Brazil.

Bruna, P. 2026. *Collective Decision-Making in Coupled Echo State Networks*. Mathematics of Intelligences Reunion Conference at Lake Arrowhead, Institute for Pure and Applied Mathematics, University of California, Los Angeles.

Bruna, P. 2025. *Collective Decision-Making in Coupled Echo State Networks*. Cognitive & Information Sciences Brownbag Series, University of California, Merced.

Bruna, P. 2025. *Collective Information Processing in Loosely Coupled Echo State Networks*. Graduate Workshop in Computational Social Science, Santa Fe Institute, New Mexico.

Bruna, P. 2025. *Basal Reservoir Computing*. Philosophy and Cognitive Science Working Group, University of California, Merced.

Bruna, P. & Gyllingberg, L. 2024. *Non-neural Pattern Completion in an Adaptive Reservoir Computer*. Mathematics of Intelligences Culminating Conference at Lake Arrowhead, Institute for Pure and Applied Mathematics, University of California, Los Angeles.

Bruna, P. 2024. *Least Effort and Alignment in Task-Oriented Communication*. Mathematics of Intelligences Seminar Series, Institute for Pure and Applied Mathematics, University of California, Los Angeles.

Bruna, P. 2024. *Concept Alignment*. Mathematics of Intelligences Working Group on Architecture and Evaluation of Generative AI, Institute for Pure and Applied Mathematics, University of California, Los Angeles.

Bruna, P. 2024. *Falandays-style Reservoir Networks*. Mathematics of Intelligences Working Group on Ecological and Homeostatic Intelligence, Institute for Pure and Applied Mathematics, University of California, Los Angeles.

Bruna, P. 2024. *Concept Alignment*. Philosophy and Cognitive Science Working Group, University of California, Merced.

Bruna, P. 2023. *Semantics and Syntax Co-emerge in Adaptive Reservoir Network Dynamics*. Diverse Intelligences Summit, University of St. Andrews, Scotland.

Bruna, P. 2020. *Hearing That It Is Silent: How to Hallucinate a Non-Perception*. 10th Annual Mid-Hudson Valley Undergraduate Philosophy Conference, Marist College, Poughkeepsie, NY. (Canceled due to COVID-19)

Contributed Posters

** Presenting author(s); ⁺ Mentored undergraduate*

*Gaskin, B., **Bruna, P.**, Jackson, I., & O'Hearn, W. (2026). *BrainlessLab: From cellular cognition to collective intelligence*. The International Conference on Artificial Life, Waterloo, Ontario, Canada.

*Nosrati, S., **Bruna, P.**, Spivey, M., & Marghetis, T. (2026). *Spatial navigation at the extremes: The collective accomplishment of high-speed, high-risk navigation in Rally Racing*. 48th Annual Conference of the Cognitive Science Society, Rio de Janeiro, Brazil.

***Bruna, P.**, Dale, R., Spivey, M., & Kello, C. 2025. *Coupled Echo State Networks as a Model of Task-Oriented Alignment*. UC Merced Cognitive Science Showcase, Merced, CA.

***Bruna, P.**, Dale, R., Spivey, M., & Kello, C. 2025. *Coupled Echo State Networks as a Model of Task-Oriented Alignment*. 47th Annual Conference of the Cognitive Science Society, San Francisco, CA.

*Gyllingberg, L., Tian, Y., Sumpter, D., & **Bruna, P.** 2025. *Oscillations and Adaptation: Mathematical Models of Non-Neural Cognition*. Biological Systems that Learn, National Institute for Theory and Mathematics in Biology, Chicago, IL.

***Bruna, P.** & Kello, C. 2024. *Alignment Through Dyadic Least Effort in Task-Oriented Conversations*. 65th Annual Meeting of the Psychonomic Society, New York City, NY.

***Bruna, P.** & Kello, C. 2024. *Alignment Through Dyadic Least Effort in Task-Oriented Conversations*. Modeling Multi-Scale Collective Intelligences, Institute for Pure and Applied Mathematics, University of California, Los Angeles.

*Kello, C. & ***Bruna, P.** 2024. *Emergent Mental Lexicon Functions in ChatGPT*. 46th Annual Conference of the Cognitive Science Society, Rotterdam, Netherlands.

*⁺Thao, K., **Bruna, P.**, & Kello, C. 2024. *Exploring Emergent Embodied Intelligence in Large Language Models: Insights from ChatGPT-4*. 15th Annual California Cognitive Science Conference, Berkeley, CA.

*Rane, S. & ***Bruna, P.** 2023. *Concept alignment*. 1st NeurIPS Workshop on AI meets Moral Philosophy and Moral Psychology (MP2), New Orleans, LA.

*⁺Leshock, E., *⁺Pena, Y., **Bruna, P.**, & Andrews, J. 2021. *Learned Categorical Perception with Artificially Generated Faces*. 36th Annual URSI Symposium, Vassar College, Poughkeepsie, NY.

*Bigler, M. & ***Bruna, P.** 2020. *Probing the Methodology and Interpretation of Learned Categorical Perception Research*. 35th Annual URSI Symposium, Vassar College, Poughkeepsie, NY.

***Bruna, P.** & *Marks, P. 2019. *Psychophysical Adaptive Procedure: Developing a New, Generalizable Method*. 34th Annual URSI Symposium, Vassar College, Poughkeepsie, NY.

Honors

2025 **Best Poster Award - 2nd place**, *Cognitive Science Showcase, University of California, Merced*
Graduate Student Highlight ("Rising Star: How Life's Journey Shaped a Cognitive Scientist's Path to Discovery"),
`https://graduatedivision.ucmerced.edu/Polyphony-J-Bruna`

2022 **Best Technical Write-Up**, *Mid-Hudson IEEE Undergraduate Capstone Project Awards*

Awards, Fellowships, & Grants

** Awarded but declined*

2026	**SSHA Dean Fellowship**, *University of California, Merced*	*$ 27,300*
	Residency Funding Award, *Institute for Humane Studies*	*$ 4,500*
	Professional Development Award, *University of California, Merced*	*$ 1,725*
	Graduate Student Travel Award, *University of California, Merced*	*$ 1,000*
	Summer Research Fellowship, *University of California, Merced*	*$ 4,050*
	Humane Studies Fellowship, *Institute for Humane Studies*	*$ 1,000*
2025	**Summer Research Fellowship**, *University of California, Merced*	*$ 5,400*
	Hayek Fund Travel Award, *Institute for Humane Studies*	*$ 3,475*
	Graduate Workshop in Computational Social Science Scholarship, *Santa Fe Institute*	*$ 1,000*
	Hayek Fund Travel Award, *Institute for Humane Studies*	*$ 2,400*
2024	**Humane Studies Fellowship**, *Institute for Humane Studies*	*$ 5,000*
	Fall Research Fellowship, *University of California, Merced*	*$ 6,000*
	MacKenzie Scott Graduate Student Travel Award, *University of California, Merced*	*$ 1,000*
	Travel Award, *University of California, Merced*	*$ 1,800*
	Summer Research Fellowship, *University of California, Merced*	*$ 5,475*
2023	**Alumni Research Award**, *Diverse Intelligences Summer Institute*	*$ 750*
	Fall Research Fellowship, *University of California, Merced*	*$ 22,500*
	Travel Award, *University of California, Merced*	*$ 800*
	Summer Research Fellowship, *University of California, Merced*	*$ 1,600*
2022	**Chancellor Recruitment Fellowship**, *University of California, Merced*	*$ 45,000*
	***Foreign Language and Area Studies (FLAS) Fellowship**, *U.S. Department of Education*	*$ 22,500*
	Adolph Sutro Fellowship, *Vassar College*	*$ 10,000*
	The Jamie Nisse Greenberg Philosophy Prize, *Vassar College*	*$ 900*
	Critical Language Scholarship for advanced Japanese, alternate, *U.S. Department of State; American Councils for International Education*	
2021	**Frances Lehman Loeb 1928 Scholarship**, *Vassar College*	*$ 44,000*
	Summer Research Fellowship, *Vassar College*	*$ 4,000*
2020	**Frances Lehman Loeb 1928 Scholarship**, *Vassar College*	*$ 44,000*
	Undergraduate Research Summer Institute Fellowship, *Vassar College*	*$ 5,000*
2019	**Frances Lehman Loeb 1928 Scholarship**, *Vassar College*	*$ 42,500*
	Undergraduate Research Summer Institute Fellowship, *Vassar College*	*$ 5,000*
2018	**1956 Memorial Scholarship**, *Vassar College*	*$ 42,000*

Teaching Experience

Spr. 2026 **Complex Adaptive Systems,** Teaching Assistant, *Cognitive Science, University of California, Merced*
Fall 2025 **Mind, Brain, and Computation,** Teaching Assistant, *Cognitive Science, University of California, Merced*
Spr. 2025 **Critical Reasoning,** Teaching Assistant, *Philosophy, University of California, Merced*
Spr. 2024 **Contemporary Moral Problems,** Teaching Assistant, *Philosophy, University of California, Merced*
Spr. 2022 **Introduction to Cognitive Science,** Teaching Assistant, *Cognitive Science, Vassar College*
Fall 2021 **Introduction to Cognitive Science,** Teaching Assistant, *Cognitive Science, Vassar College*

Outreach & Professional Development

Workshops & Training

2026 **Santa Fe Institute,** Advanced Graduate Workshop in Computational Social Science, *Santa Fe, New Mexico*
Diverse Intelligences Summer Institute, *Geneva, NY*

2025 **Santa Fe Institute,** Graduate Workshop in Computational Social Science, *Santa Fe, New Mexico*
Complexity Science Hub, Winter School on Complex Systems in Our Changing World, *Vienna, Austria*

2024 **Institute for Pure and Applied Mathematics,** Modeling Multi-Scale Collective Intelligences, *University of California, Los Angeles*
Berggruen Institute, Future Humans Workshop on Computational Consciousness, *Beverly Hills, California*
Institute for Pure and Applied Mathematics, Naturalistic Approaches to Artificial Intelligence, *University of California, Los Angeles*
Institute for Humane Studies, Aristotle and Human-Centered AI, *Webinar*
Institute for Pure and Applied Mathematics, Theory and Practice of Deep Learning, *University of California, Los Angeles*
Dynamic Interactions and Methodologies Symposium, *University of California, Los Angeles*
Institute for Pure and Applied Mathematics, Analyzing High-dimensional Traces of Intelligent Behavior, *University of California, Los Angeles*

2023 **Sante Fe Institute,** Complexity-GAINs International Summer School on Intelligence and Representation, *Isaac Newton Institute for Mathematical Sciences, University of Cambridge*
Diverse Intelligences Summer Institute, *University of St. Andrews*

2022 **Phi Beta Kappa,** Key Connections, *Webinar*

2020 **Center for Philosophy of Science,** Pittsburgh Summer Program in Philosophy of Science and Specialized Sciences, *University of Pittsburgh and Carnegie Mellon University*

Service Positions

2025-cont. **Graduate Book Club President,** Cognitive and Information Sciences, *University of California, Merced*
Merced Complexity Lab Co-Lead, Cognitive and Information Sciences, *University of California, Merced*

2025 **Brownbag Organizer,** Cognitive and Information Sciences, *University of California, Merced*

2023-2024 **Graduate Student Chair,** Cognitive and Information Sciences, *University of California, Merced*
Research Mentor, Learning Aligned Employment Program, *University of California, Merced*

2021 **Research Mentor,** Undergraduate Research Summer Institute, *Vassar College*

2018-2019 **Editor,** Vassar College Journal of Philosophy, *Vassar College*

Peer Review

Cognition
Cognitive Science
Journal of Experimental Psychology: General
Minds and Machines
Journal of Supercomputing
Proceedings of the Annual Meeting of the Cognitive Science Society

Skills

Programming:	Python, R, JavaScript, HPC parallel computing
Open source software:	crqa (R package, contributor)
Languages:	English (native), Japanese (basic)